Mohammad Askari

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EDUCATION

UNIVERSITY OF CALIFORNIA, LOS ANGELES (UCLA)

PhD Candidate in Civil Engineering — cGPA 4.0

MSc Candidate in Statistics — cGPA 3.9

TECHNICAL EXPERIENCE

UCLA Los Angeles, CA — Sep 2021 - Present

Graduate Student Researcher

- Lead developer of CHEERSafe, a parallelized Python library (11K lines, 500+ tests) estimating hurricane wind and flood losses for 1.15M single-family homes in 45 North Carolina counties; damage-library generation runs 65× faster on 112 cores. [🔗]
- Built a linear-programming calibration (Gurobi) that assigns annual probabilities to 932 hurricane scenarios to match 5,400 ASCE-7 wind targets and FEMA flood-zone targets for the NSF CoPe-CHEER project.
- Implemented North Carolina's AI-driven building-inventory pipeline (5.65M buildings from FEMA, Microsoft, NSI, and parcel data) with a ViT roof-shape classifier (F1 0.92) and gradient-boosting imputation cutting structure-value error by 37%.
- Co-first author of RAPID-A (*ASCE OPEN*, 2025), an uncertainty-aware ensemble of deep segmentation models that assesses post-earthquake building damage from satellite images of the 2023 Türkiye earthquakes and generalizes to unseen cities.
- Co-developed an XGBoost claim-occurrence model (736K building–hurricane observations, 4 hurricanes) that cut total claim-count error to 5.6% from 42.1% with logistic regression on held-out data; 2nd author, *Natural Hazards* (2026).
- Developed transformer-based NLP and ML techniques for PII removal and text classification of damage descriptions for the North Carolina Insurance Underwriting Association (NCIUA).
- Rebuilt ground-motion record selection in JAX as a GPU-resident greedy search: 2,000×+ faster than the legacy OpenCL tool [🐙] (122 s → 0.06 s for 200 records) with equal or better mean-spectrum fit.

OPAL-AI Los Angeles, CA — Oct 2024 - Jan 2025

Machine Learning Engineer (Volunteer)

Astro Adjuster Project [🔗]

- Developed an *agentic* Vision–Language Model (VLM) pipeline (LangChain, Google Cloud Vertex AI) turning site-captured video into structured claims-inspection reports for Astro Adjuster, which cuts claims reporting from hours to minutes.
- Engineered a retrieval-augmented generation (RAG) stack with LlamaIndex (vector indexing, chunking, semantic retrieval) delivering citation-grounded Q&A over the generated reports.

NASA SBIR Ignite: FireGPT Project [🔗]

- Contributed as an ML engineer to Opal AI's winning NASA SBIR Ignite Phase I proposal for *FireGPT*, a vision–language model for prescribed-burn decision support ($166K Phase I; $1.27M follow-on Phase II).

PROJECTS (SELECTED)

Open-source and graduate-level

- **Overleaf Web MCP** (open source; TypeScript, Node.js): Built and shipped a Model Context Protocol (MCP) server with 31 tools that lets AI assistants (Claude Code, Codex, Cursor) read, edit, compile, and review Overleaf projects through the user's own session; 2.3K+ npm downloads in three months. Engineered safeguards for agent-driven edits: revision-checked writes, tracked changes, and plan-then-confirm folder sync. [🐙]
- **Proper Scoring Rules to Reduce LLM Overconfidence** (Python, LLM evaluation): Demonstrated that stating the logarithmic scoring rule in the prompt curbs DeepSeek-V4-Flash's overconfidence: +0.54 nats in log score on MMLU and +1.49 on MMLU-Pro (1,000 questions each) at negligible accuracy cost. [🐙]

- **Causal Reasoning Agent** (team of 4; Python): Co-developed an LLM-agnostic agent framework with Kripke epistemic reasoning; built its Mastermind and 2048 environments, evaluation runners, and a decision-trace dashboard. [GitHub]
- Built an efficient ML pipeline that optimizes XGBoost hyperparameters via experimental design techniques, outperforming grid search in predicting wildfire damage to buildings. — Advisor: Professor Hongquan Xu
- Implemented Deep Q-learning with experience replay (DQN) and Deep Deterministic Policy Gradient (DDPG) reinforcement learning algorithms for Racing Car and Double Pendulum problems. — Advisor: Professor Jonathan Kao
- Evaluated performance of CRNN, LSTM, conditional GANs, and CNN models on time series data using TensorFlow, highlighting robust deep learning methods for signal processing. — Advisor: Professor Jonathan Kao [GitHub]
- A tutorial on the doubly robust targeted maximum likelihood (TMLE) estimation method and its implementation in R, merging causal inference with ML principles for improved estimation. — Advisor: Professor Chad Hazlett

TECHNICAL SKILLS

Programming Languages: Python, C++, R, TypeScript, SQL, Bash, MATLAB
ML & DL Frameworks: PyTorch, JAX, TensorFlow, Keras, Scikit-learn, XGBoost, Hugging Face Transformers, spaCy
LLMs & Agents: LangChain, LlamaIndex, RAG, MCP, Google Cloud Vertex AI
Tools: Git, GitHub, CI/CD, Docker, OpenCL, Gurobi

PUBLICATIONS AND PRESENTATIONS

Articles

[1] N. O. Tackie-Otoo, **M. Askari**, P. Hadinata, R. A. Davidson, E. Taciroglu, and G. Hardy, "Hurricane wind loss modeling using insurance claims data," *Natural Hazards*, vol. 122, no. 7, p. 300, 2026. doi: 10.1007/s11069-026-08059-z.

[2] **M. Askari**, M. H. Soleimani-Babakamali, C. Williams, R. Davidson, and E. Taciroglu, *CHEERSafe: A Python Library to Assess Hurricane Damage and Loss*, Manuscript in preparation, 2026.

[3] M. H. Soleimani-Babakamali, **M. Askari**, M. A. Heravi, R. Sisman, N. Attarchian, A. Askan, R. Soleimani, and E. Taciroglu, "Deep Ensemble Learning for Rapid Large-Scale Postearthquake Damage Assessment: Application to Satellite Images from the 2023 Türkiye Earthquakes," *ASCE OPEN: Multidisciplinary Journal of Civil Engineering*, vol. 3, no. 1, p. 04 025 003, 2025. doi: 10.1061/AOMJAH.AOENG-0043.

[4] S.-Z. Chen, X.-Y. Wang, X.-R. Ma, T. Wang, **M. Askari**, and E. Taciroglu, "Transforming Regression to Classification for Enhancing Extrapolation Capacity of Data-driven Structural Behavior Prediction," *Expert Systems with Applications*, vol. 277, p. 127 302, 2025, issn: 0957-4174. doi: https://doi.org/10.1016/j.eswa.2025.127302.

[5] R. Soleimani-Babakamali, M. H. Soleimani-Babakamali, M. A. Heravi, **M. Askari**, O. Avci, and E. Taciroglu, "Transferring Damage Detection Knowledge Across Rotating Machines and Framed Structures: Harnessing Domain Adaptation and Contrastive Learning," *Mechanical Systems and Signal Processing*, vol. 221, p. 111 743, 2024. doi: https://doi.org/10.1016/j.ymssp.2024.111743.

[6] M. Shokrabadi, Y. Bozorgnia, H. V. Burton, J. W. Baker, and **M. Askari**, "An Efficient Computational Platform for Selecting and Scaling Ground Motion Records while Considering Multiple Target Spectra," University of California, Los Angeles, Tech. Rep., 2022. doi: https://doi.org/10.34948/N3K01N.

[7] **M. Askari** and M. Mahsuli, "Modeling of Epistemic Uncertainties in Seismic Hazard Analysis Based on Reliability Methods," Center for Infrastructure Sustainability and Resilience Research, Sharif University of Tech., Tech. Rep., 2020.

Presentations

May 2025	EMI Conference	CHEERSafe: A Modular Object-Oriented Framework for Regional Hurricane Risk Assessment
May 2024	PMC/EMI Conference	An End-To-End Framework for Building Inventory Generation Through Public Data Fusion and Artificial Intelligence
Feb 2024	NHERI Computational Symposium	Deep Ensemble Learning for Rapid Large-Scale Post-Earthquake Damage Assessment—Application to 2023 Turkiye Earthquakes Satellite Images (*poster*)
Feb 2022	ASCE Lifelines Conference	An Efficient Computational Platform for Selecting and Scaling Ground Motion Records while Considering Multiple Target Spectra (*poster*)

CERTIFICATIONS

Jul 2024	TACC	AI Computational Academy & Hackathon
Jan 2024	Coursera	AI for Good Specialization [🔗]
Nov 2023	NVIDIA	Building Transformer-Based NLP Applications [🔗]
May 2023	Coursera	Deep Learning Specialization [🔗]
Feb 2021	Coursera	Machine Learning [🔗]

COMMUNITY ENGAGEMENT & VOLUNTEER EXPERIENCE

Sep 2024-present	Elsevier Engineering Structures ASCE Open ASCE I3CE	Journal article reviewer related to ML and AI applications
2024–2025	LA Hacks	Expert judge for the LA Hacks hackathon
Feb 2024-present	NSF	Code and documentation management of CoPe-CHEER NSF project
Sep 2022-July 2023	UCLA	Board member of the Iranian Graduate and Professional Association, IGPA